SEAT NO. CT-24020

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(Computer Science)

FALL SEMESTER EXAMINATIONS 2025

Time : 3 Hours | **Batch 2024** | **Dated : 11-DEC-25** | **Max Marks : 60**

Probability & Statistics - MT-331

INSTRUCTIONS:

- *Attempt All questions.*
- *Statistical Annexure is attached on last page.*
- *Use of scientific calculator is allowed.*

CLO-1

Q.1 A network provider investigates the load of its network. The number of concurrent users is recorded at 31 locations (thousands of people), 12

17.2 22.1 18.5 17.2 18.6 14.8 21.7 15.8
24.1 13.3 16.2 17.5 19.0 23.9 14.8 22.2
13.5 15.8 13.1 16.1 21.9 23.9 19.3 12.0
15.4 16.7 19.5 16.2 16.9 17.1 20.2

a. **Estimate** the sample mean, variance, and standard deviation of the number of concurrent users?
b. **Locate** the five-point summary and construct a box plot?
c. Compute the interquartile range. Are there any outliers?

CLO-1

Q.2(a) A computer manager needs to know how efficiency of her new computer program depends on the size of incoming data. Efficiency will be measured by the number of processed requests per hour. Applying the program to data sets of different sizes, she gets the following results, 08

Data size (gigabyte), x	6	7	7	8	10	10	15
Processed requests, y	40	55	50	41	17	26	16

In general, larger data sets require more computer time and therefore, fewer requests are processed within 1 hour. The response variable here is the number of processed requests (y), and we attempt to predict it from the size of a data set (x).

i. Fit a linear regression model estimating the processed requests (y) to predict it from the size of a data set (x).

$$y = a + bx$$

ii. **Estimate** the processed requests, y when the size of a data set is 12 gigabytes?
iii. **Estimate** coefficient of correlation and comments?

(b) A computer assembling company receives 24% of parts from supplier X, 36% of parts from supplier Y, and the remaining 40% of parts from supplier Z. Five percent of parts supplied by X, ten percent of parts supplied by Y, and six percent of parts supplied by Z are defective. If an assembled computer has a defective part in it, **Estimate** the probability that this part was received from supplier Z? 04

P.T.O

CamScanner

CLO-2

Q.3(a) A clinic offers a weight-loss program. A review of its records found the following amounts of weight loss, in pounds, for a random sample of 24 of its clients at the conclusion of a 4-month program: **06**

18	25	16	11	15	20	16	19
28	25	26	31	45	40	36	19
28	25	36	16	35	20	16	19

a. **Calculate** a 95% confidence interval for the population mean.

b. At the 5% significance level, **apply** the appropriate statistical test to assess whether the average amount of weight loss is less than 26.

(b) A hard drive's read speed is normally distributed with mean 120 MB/s and SD 10 MB/s. **06**

i. **Calculate** the probability that a randomly selected hard drive has a read speed greater than 130 MB/s?

ii. If a random sample of 25 hard drives is taken from the same population, **Calculate** the probability that the sample mean read speed is greater than 124 MB/s?

CLO-2

Q.4(a) A hardware tester finds a defective chip with probability of 0.2. **Calculate** the probability that the **06**

i. 4th defective chip is found on the 10th trial.

ii. 1st defective chip is found on the 4th trial.

(b) A hard disk develops bad sectors at a rate of 0.5 per GB. **Calculate** the probability that **06**

i. A 2 GB area has at least 2 bad sectors.

ii. A 4 GB area has at most 2 bad sectors.

CLO-2

Q.5(a) A box contains 8 working USB drives and 2 defective ones. If 3 are selected, **Calculate** the probability that **06**

i. All are working.

ii. Exactly 1 is not working.

(b) A quality control engineer tests the quality of produced computers. Suppose that 25% of computers have defects, and defects occur independently of each other. **Calculate** the probability of **06**

i. Exactly 3 defective computers in a shipment of twenty.

ii. At least 3 defective computers in a shipment of twenty.

----END----

CamScanner